1. Introduction

The landscape of machine learning has undergone rapid transformation in recent years. With generative AI capturing headlines and organizations racing to adopt large language models, one might assume that classical algorithms have been relegated to the sidelines. Yet industry data tells a more nuanced story. This report investigates which machine learning models are most widely used in real industrial settings as of 2025–2026, drawing on industry surveys, Kaggle competition data, company technical blogs, and market analyses. It then offers a forward-looking assessment of how the distribution of widely used models will evolve over the next 5–10 years.

2. Current State: Most Widely Used Models in Industry

2.1 The Dominant Family: Gradient Boosting Machines

Gradient Boosting Machines (GBM), particularly XGBoost, LightGBM, and CatBoost, have emerged as the most widely used and practically dominant family of models in industrial settings today. This finding is supported by multiple independent sources:

- **Kaggle competition dominance.** Since its introduction in 2016, XGBoost has become a fixture in winning solutions across predictive modeling competitions, especially for structured data problems. Gradient boosting models consistently outperform alternatives on tabular data – the dominant data format across most business domains.
- **Superior performance across industry benchmarks.** Studies comparing XGBoost against logistic regression, random forests, and neural networks consistently find gradient boosting as the top performer. In credit risk prediction – a core banking application – XGBoost achieved the highest accuracy (88.7%), precision (89.5%), recall (80.3%), and AUC (91.3%) among all models tested. Similarly, in customer churn prediction for the software industry, XGBoost demonstrated superior recall (0.85) and ROC AUC (0.86) compared to logistic regression.
- **Real-time and adaptive capabilities.** Industry analysts report that gradient boosting frameworks are actively evolving to support real-time and incremental learning, eliminating the need for costly full retraining and making them even more attractive for dynamic production environments.

The 2026 industry standard top five includes Linear/Logistic Regression, Random Forests, Support Vector Machines (SVM), Gradient Boosting Machines (XGBoost/LightGBM), and Artificial Neural Networks – with gradient boosting singled out for "top-tier tabular data performance".

2.2 The Persistent Role of Classical Models

Despite the dominance of gradient boosting, **simpler classical models – particularly linear and logistic regression – remain extensively used in industry**, albeit for different purposes. Linear and logistic regression serve as "baseline powerhouses," providing highly interpretable, low-latency solutions for straightforward prediction tasks. Their explainability is a critical advantage in regulated industries: financial institutions and healthcare organizations facing regulatory scrutiny often prefer linear models that can be transparently audited.

2.3 The Rise of Large Language Models

Large language models have achieved remarkably rapid enterprise adoption, but this adoption is largely additive rather than substitutional. According to KPMG's 2025 Cloud Monitor survey of 509 companies, 92% of respondents use LLMs in their daily work or have implemented initial LLM integrations into business processes. OpenAI's GPT leads with 61% usage, followed by Google's Gemini at 40%, while 26% of companies use open-source solutions primarily for digital sovereignty reasons.

However, a critical nuance emerges: LLMs and traditional ML models are not competing head-to-head across all use cases. As an MIT Sloan analysis notes, while generative AI is widely accessible and useful, businesses need to know when to use other AI tools like traditional machine learning. Dedicated models achieve prediction accuracy rates of 91.2% while requiring only 34% of the computational resources needed for larger, general-purpose models. For tasks requiring speed, precision, and control, optimized traditional models have consistently outperformed LLMs in cost and latency, while GenAI excels in more open-ended, fuzzy scenarios.

MODEL FAMILY	PRIMARY USE CASES	ADOPTION STRENGTH
GRADIENT BOOSTING (XGBOOST, LIGHTGBM, CATBOOST)	Tabular data regression & classification	**Highest** for structured data
LINEAR/LOGISTIC REGRESSION	Baseline predictions, low-dimensional problems	**High** (interpretability driven)
RANDOM FORESTS	Robust classification, ensemble baseline	**High** (simplicity & robustness)

DEEP LEARNING (CNN, RNN, TRANSFORMERS)	Computer vision, NLP, sequence modeling	**High** for unstructured data
LARGE LANGUAGE MODELS	Text generation, summarization, chatbots	**Very high** (92% of companies)

3. Future Outlook: How Will Model Distribution Change Over the Next 5–10 Years?

The next decade will likely see a **coexistence and hybridization rather than wholesale replacement** of classical approaches. Three major forces will shape the evolution:

3.1 Classical Models Will Remain Dominant for Structured Data

Gradient boosting and classical models will maintain their stronghold on tabular data problems. This prediction is grounded in empirical evidence: large-scale comparative studies consistently show gradient-boosted decision trees outperforming deep learning models on tabular data, for reasons that have been carefully analyzed in recent research (Grinsztajn et al., 2022; Shwartz-Ziv and Armon, 2022). As the UW-Madison Data Science program notes, XGBoost often outperforms deep learning on small to medium-sized datasets, offers faster training times, and provides greater interpretability – all critical factors for production deployment.

The financial services sector exemplifies this enduring relevance. Despite significant investment in deep learning research, quantitative finance and risk management teams continue to rely heavily on tree-based models like XGBoost, which outperform deep learning architectures in core quant finance tasks.

3.2 LLMs and Generative AI Will Gain Share – But in Complementary Domains

LLM adoption will accelerate further and expand into new application areas, but traditional ML will not be displaced where it excels. The Enterprise Large Language Model Market, valued at USD 11.25 billion in 2025, is projected to reach USD 60.52 billion by 2032, a CAGR of 27.15%. Enterprise leaders expect an average of 75.7% growth in LLM spending over the next year, from $7 million in May 2025 to $12.3 million on average in 2026.

However, this growth will largely occur in domains suited to LLM capabilities: unstructured text processing, conversational interfaces, code generation, summarization, and creative tasks. By contrast, high-stakes predictive applications requiring precision, low latency, and provable behavior – fraud detection, credit scoring, medical diagnosis, real-time recommendation engines – will remain the domain of traditional ML models. The KPMG survey reinforces this complementarity: 69% of LLM users pursue a multi-model approach instead of committing to a single provider, suggesting a pragmatic recognition that no single model class is optimal for all tasks.

3.3 Hybrid Architectures Will Emerge as the Norm

The strongest shift over the next decade may not be from one model class to another, but toward **hybrid systems that combine strengths across families**. Industry is already moving away from the question "which model is best?" toward "how can multiple models work together effectively?" This trend is reflected in the rapid growth of the MLOps market, which enables organizations to manage increasingly complex multi-model production pipelines. The MLOps market was valued at USD 2.33 billion in 2025 and is projected to reach USD 25.93 billion by 2034, a CAGR of 28.9%.

3.4 Domain-by-Domain Shift Analysis

Finance, insurance, and healthcare – where interpretability, regulatory compliance, and precision are paramount – will be the slowest to shift away from classical models. In these sectors, the cost of a false prediction is high and the need for auditability is non-negotiable. Gradient boosting and logistic regression will remain standard tools.

Technology companies, marketing, and creative industries will see the most dramatic shift toward LLMs and generative AI. Content generation, customer service automation, code assistance, and creative workflows are natural fits for generative models. The technology sector has already seen substantial adoption, and this will accelerate as small language models (SLMs) become more efficient and capable of running on edge devices.

Manufacturing, logistics, and IoT applications will see increased adoption of specialized deep learning architectures for computer vision, predictive maintenance, and time-series forecasting, while continuing to rely on traditional models for core operational analytics.

4. Conclusion

The evidence from industry surveys, Kaggle competition trends, and market analyses points to a clear conclusion: **there is no single "most widely used" model family, but rather a stratified landscape where gradient boosting machines represent the current workhorse for tabular data problems, while classical linear models and deep learning architectures occupy distinct niches based on data type, interpretability needs, and computational constraints.** Large language models have achieved extraordinary adoption velocity, but thus far have expanded the total addressable AI market rather than cannibalizing traditional ML deployments.

Over the next 5–10 years, this stratification will persist but become more nuanced. The "Cambrian explosion" of model architectures will give way to a more mature ecosystem where organizations routinely deploy multiple model types in hybrid pipelines. Classical models will not be displaced from their strongholds on structured, tabular, and high-stakes predictive tasks

– not because of inertia, but because they remain genuinely superior in those domains. The future of industrial machine learning is not replacement, but orchestration.

References

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